Marco Del Fava

Neural Signal Processing & Acquisition

Biomedical Engineering M.S. | UC Davis | Human Intracranial Electrophysiology

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Research Profile

Developing and validating software for human intracranial electrophysiology, including multichannel neural acquisition, signal processing, spike sorting, and experimental synchronization. Experienced in Python-based neural data pipelines, acquisition hardware integration, and quantitative evaluation against recorded data.

Technical Skills

Neural Data & Signal Processing

- SpikeInterface, Kilosort4, MountainSort5; waveform features and unit matching
- Offline waveform discrimination and afferent clustering

Neural Acquisition & Tooling

- Intan RHD, Neuralynx CSC, MED; probe mapping and acquisition configuration
- DHN-AQ RC/CS configuration, SSH orchestration, and NRD replay prototypes

Machine Learning & Data Science

- Python (NumPy, Pandas, scikit-learn, PyTorch, Numba); MATLAB
- Sentence embeddings and seeded NEURON simulation prototypes

General Technical

- Git, Linux, SSH, shell scripting, Docker, and AWS GPU workflows
- Microsoft 365 administration; SQL and PostgreSQL

Selected Engineering Projects

Neural Acquisition and Synchronization — Implemented

- Built Intan RHD ingestion, probe/channel checks, and SpikeInterface/Kilosort4 processing with waveform and quality-metric extraction.
- Implemented queued PsychoPy/RHX event synchronization, exact-label joins, and affine clock fitting to analyze clock drift and alignment residuals.

Acquisition workflow and clock-alignment software implemented; timing performance not independently verified.

Intan processing notebook

Neural Signal Fidelity and Processing Validation — Validated

- Evaluated DarkHorse Neuro compression on 20 million samples (one channel, 32 kHz; 625 s); maximum source-to-MED difference was 0.5 uV, with zero mismatches after rounding source values to whole microvolts.
- Controlled MountainSort5 comparison of rounded inputs preserved all 35,587 spike frames for the compared unit (exact Jaccard 1.0).

Offline validation on a bounded single-channel slice; archived outputs reviewed, not a new raw-data rerun.

Waveform fidelity evidence · Controlled spike comparison

Spike Sorting and Waveform Discrimination — Validated

- Implemented waveform-feature evaluation across 38,145 spike snippets and 10 putative O-Sort unit labels; event-family KNN purity at k=5 was 95.04%.
- Measured 54.26% Hungarian-matched KMeans accuracy (ARI 0.3665) for that feature family; neighborhood purity is not sorter accuracy or biological ground truth.

Offline feature validation; streaming detection and competitive-learning sorting in SNNeuro remain prototypes.

Discrimination results · O-Sort membership comparison · Streaming prototype

Single-neuron analysis and adaptive-search development — Prototype

- Implemented a seeded NEURON prototype mapping sentence similarity to stochastic input rates, with fixed synaptic weights.
- Added seeded stimulus windows, presynaptic event recording, voltage traces, and output spike times for reproducible simulated response analysis.

Simulation implemented; adaptive human stimulus selection and semantic selectivity are not validated.

Simulation implementation

Supporting acquisition work - Neuralynx / Dark Horse Neuro — Prototype

- Implemented DHN-AQ RC/CS configuration writers and SSH launch orchestration.
- Developed an NRD-to-UDP replay prototype for DHN-AQ ingestion.

Prototype implemented; hardware reception and recording validation remain in progress.

Acquisition implementation

Research Experience

Research Volunteer, UC Davis Health — Z-Lab (Dr. Jie Zheng) — Jul 2025 - Present

- Develop acquisition and spike-processing workflows with probe mapping, waveform extraction, quality metrics, and experimental clock alignment; distinguish exploratory stimulus responses from semantic tuning.
- Analyze human intracranial single units with O-Sort/SpikeInterface; develop narrative-aligned rasters and peri-stimulus firing-rate workflows.

Education

M.S. Biomedical Engineering, University of California, Davis — Fall 2026 - Present

Python for Data Science, Professional Concentration, UC Davis Continuing and Professional Education — 2025

M.S. Psychology, Arizona State University — 2023

B.S. Neuroscience, Physiology, and Behavior, University of California, Davis — 2017

Additional Professional Experience

IT Administrator, First Step Communities — Jul 2021 - Present

- Administer multi-site networks, routing/VLAN segmentation, Microsoft 365/Exchange Online, and camera/recording systems; troubleshoot infrastructure and support longitudinal service-utilization and housing-outcome research.

Analyst, Eurofins Test America — Oct 2019 - Jun 2021

- Supported analytical laboratory QA, data review, and standard operating procedures.

Certifications

- Data Analysis with Matplotlib, Pandas, and Scikit-Learn Python Libraries — UC Davis Continuing and Professional Education (2025-09-18)
- Biomedical Research; Research Administration; GCP Social and Behavioral Research — CITI Program